

D 1 A

$$A = \{x \in \mathbb{R} \mid x^2 - 2x \geq 1\} = (-\infty, 1 - \sqrt{2}] \cup [1 + \sqrt{2}, +\infty)$$

$$x^2 - 2x - 1 \geq 0$$

$$\Delta = b^2 - 4ac = 4 - 4(1)(-1) = 4 + 4 = 8$$

$$x_{1/2} = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

$$x_1 = 1 - \sqrt{2} \quad x_2 = 1 + \sqrt{2}$$

D 1 B

$$A = \{x \in \mathbb{R} \mid x^2 - 6x \leq 9\} = [3 - 3\sqrt{2}, 3 + 3\sqrt{2}]$$

$$x^2 - 6x - 9 \leq 0$$

$$\Delta = b^2 - 4ac = 36 - 4(1)(-9) = 36 + 36 = 72$$

$$x_{1/2} = \frac{6 \pm \sqrt{72}}{2} = \frac{6 \pm \sqrt{6^2 \cdot 2}}{2} = \frac{6 \pm 6\sqrt{2}}{2} = 3 \pm 3\sqrt{2}$$

$$x_1 = 3 - 3\sqrt{2} \quad x_2 = 3 + 3\sqrt{2}$$

D 4 A

$$y = \begin{cases} e^{1/x} + \cos x & x < 0 \\ x^2 + e e^x & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} e^{1/x} + \cos x = e^{1/0^-} + \cos 0^- = e^{-\infty} + 1 = 0 + 1 = 1$$

$$\lim_{x \rightarrow 0^+} x^2 + e e^x = 0 + e e^0 = 0 \cdot 1 = 0$$

for for x the y is continue in $x=0$, $0=1$

D 4 B

$$y = \begin{cases} x^2 - e e^x & x \leq 0 \\ \sin\left(x + \frac{\pi}{2}\right) + e^{-1/x} & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} x^2 - e e^x = 0 - e e^0 = -e$$

$$\lim_{x \rightarrow 0^+} \sin\left(x + \frac{\pi}{2}\right) + e^{-1/x} = \sin\left(\frac{\pi}{2}\right) + e^{-1/0} = 1 + e^{-\infty} = 1$$

for for x the y is continue in $x=0$, $0=-1$

6 A

$$y = 3x^3 - 6x^2$$

$$y' = 9x^2 - 12x$$

$$y'' = 18x - 12 = 6(3x - 2)$$

$$3x - 2 < 0 \Rightarrow 3x < 2 \Rightarrow x < 2/3$$

6 B

$$y = 2x^2 - 4x^3$$

$$y' = 4x - 12x^2$$

$$y'' = 4 - 24x = 4(1 - 6x)$$

$$1 - 6x > 0 \Rightarrow -6x > -1 \Rightarrow 6x < 1 \Rightarrow x < 1/6$$

DSA

$$y = \frac{x}{\sqrt{x-1}}$$

$$y' = \frac{(\sqrt{x-1}) - (x) \left(\frac{1}{2\sqrt{x-1}} \right)}{x-1}$$

$$y'(2) = \frac{(\sqrt{2-1}) - (2) \left(\frac{1}{2\sqrt{2-1}} \right)}{1} = \frac{(1) - (2) \left(\frac{1}{2} \right)}{1} = \frac{1-1}{1} = 0$$

DSB

$$y = \frac{x}{\sqrt{x-2}}$$

$$y' = \frac{(\sqrt{x-2}) - (x) \left(\frac{1}{2\sqrt{x-2}} \right)}{x-2}$$

$$y'(3) = \frac{(\sqrt{3-2}) - (3) \left(\frac{1}{2\sqrt{3-2}} \right)}{3-2} = \frac{(1) - (3) \left(\frac{1}{2} \right)}{1} = 1 - 3/2 = \frac{2-3}{2} = -\frac{1}{2}$$

D 8 A

$$\int_0^2 \frac{x}{x+1} = \left[x+1 - \ln(x+1) \right]_0^2 = (3 - \ln(3)) - (1 - \ln(1)) = (3 - \ln(3)) - (1) = 3 - \ln(3) - 1 = 2 - \ln(3)$$

$$\int \frac{x}{x+1} dx = \int \frac{x+1-1}{x+1} dx = \int \frac{y-1}{y} dy = \int 1 - \frac{1}{y} dy = \int 1 dy - \int \frac{1}{y} dy = y - \ln(y) = x+1 - \ln(x+1) + C$$

$$y = x+1$$

$$dy = dx$$

D 8 B

$$\int_{-1}^1 \frac{x}{x+2} dx = \left[x+2 - 2 \ln(x+2) \right]_{-1}^1 = (1+2 - 2 \ln(1+2)) - (-1+2 - 2 \ln(-1+2)) = (3 - 2 \ln(3)) - (1 - 2 \ln(1)) = (3 - 2 \ln(3)) - 1 = 2 - 2 \ln(3)$$

$$\int \frac{x}{x+2} dx = \int \frac{x+2-2}{x+2} dx = \int \frac{y-2}{y} dy = \int 1 - \frac{2}{y} dy = \int 1 dy - 2 \int \frac{1}{y} dy = y - 2 \ln(y) = x+2 - 2 \ln(x+2) + C$$

$$y = x+2$$

$$dy = dx$$

D 3 A

$$\lim_{n \rightarrow +\infty} n^2 \sin\left(\frac{3}{n}\right) = n^2 \cdot \frac{\sin\left(\frac{3}{n}\right)}{\frac{3}{n}} \cdot \frac{3}{n} = n^2 \cdot \frac{3}{n} = 3n = +\infty$$

D 3 B

$$\lim_{n \rightarrow +\infty} 3n^2 \ln\left(1 - \frac{1}{n}\right) = 3n^2 \cdot \frac{\ln\left(1 - \frac{1}{n}\right)}{-\frac{1}{n}} \cdot -\frac{1}{n} = 3n^2 \cdot -\frac{1}{n} = -3n = -\infty$$

D 7 A

$$y = \cos x - 2 \ln(1+x) + 2x$$

$$y' = -\sin x - \frac{2}{1+x} + 2 \quad y'(0) = -\sin(x) - 2 + 2 = 0$$

$$y'' = -\cos x + \frac{2}{(1+x)^2} \quad y''(0) = 1 + 2 = 3$$

D 7 B

$$y = \ln(1+x) - \sin(x) - x^2$$

$$y' = \frac{1}{1+x} - \cos x - 2x \quad y'(0) = 1 - 1 = 0$$

$$y'' = -\frac{1}{(1+x)^2} + \sin x - 2 \quad y''(0) = -1 - 2 = -3$$

D 2 A

$$\sum_{n=1}^{+\infty} \frac{3 \ln(n) + 2n^2}{5n^3 \sqrt{n}} \sim \frac{2n^2}{5n^3 \sqrt{n}} = \frac{2}{5n^{3/2}}$$

$$\frac{2}{5n^{3/2}} \cdot n^{3/2} = \frac{2}{5} \in (0, +\infty)$$

$\frac{1}{n^{3/2}}$ converge quindi anche n converge

D 2 B

$$\sum_{n=1}^{+\infty} \frac{2 \ln n + 3n^3}{7n^3 \sqrt{n}} \sim \frac{3n^3}{7n^3 \sqrt{n}} = \frac{3}{7\sqrt{n}}$$

Infine per chi non soddisfa le condizioni di convergenza

Eg 1A

$$y = x^2 - 3x - 2 \lg(x) + 2 + 2 \lg(2)$$

DOMINIO

$$D = (0, +\infty)$$

LIMITI

$$\lim_{x \rightarrow 0^+} x^2 - 3x - 2 \lg(x) + 2 + 2 \lg(2) = 0 - 0 - \infty + 2 + 2 \lg(2) = -\infty \quad \text{Ass. verticale}$$

$$\lim_{x \rightarrow +\infty} x^2 - 3x - 2 \lg(x) + 2 + 2 \lg(2) \sim x^2 = +\infty \quad \text{Ass. obliquo}$$

$$\lim_{x \rightarrow +\infty} \frac{x^2 - 3x - 2 \lg(x) + 2 + 2 \lg(2)}{x} \sim \frac{x^2}{x} = x = +\infty \quad \text{no Ass. obliquo}$$

MONOTONIA

$$y = x^2 - 3x - 2 \lg(x) + 2 + 2 \lg(2)$$

$$y' = 2x - 3 - \frac{2}{x}$$

$$2x - 3 - \frac{2}{x} > 0 \Rightarrow \frac{2x^2 - 3x - 2}{x} > 0 \quad \begin{cases} x < -1/2 \\ x > 2 \end{cases} \quad \begin{cases} x > 2 \\ x > 0 \end{cases} \Rightarrow x > 2$$

y crescente per $(2, +\infty)$ e decrescente per $(0, 2)$

MAX & MIN

$$\frac{2x^2 - 3x - 2}{x} = 0 \quad x_1 = -1/2 \quad x_2 = 2$$

$$y(2) = 4 - 6 - 2 \lg(2) + 2 + 2 \lg(2) = 0$$

$x=2$ è punto di minimo assoluto ed assume valore 0

sapendo che il minimo globale della funzione è 0 per $x=2$ $y > 0$ per $x \in (0, 2) \cup (2, +\infty)$

D 3 A

$$f(x) = x^2 \sin(x)$$

1)

$$\int x^2 \sin(x) = -x^2 \cos x - \int 2x \cos x = -x^2 \cos x - 2 \int -x \cos x = -x^2 \cos x - 2 \left(-x \sin x - \int -\sin x \right) = -x^2 \cos x - 2(-x \sin x + \cos x)$$

$$-x^2 \cos x - 2(x \sin x + \cos x) = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

$$F(x) = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

$$F(0) = 2 + C \Rightarrow C = -2$$

2)

$$F(0) = -2$$

$$F'(0) = 0$$

$$F'' = (2x)(\sin x) + (x^2)(\cos x) = 2x \sin x + x^2 \cos x \quad F''(0) = 0$$

$$F''' = (2)(\sin x) + (2x)(\cos x) + (2x)(\cos x) + (x^2)(-\sin x)$$

$$2 \sin x + 2x \cos x + 2x \cos x - x^2 \sin x$$

$$2 \sin x + 4x \cos x - x^2 \sin x \quad F'''(0) = 0$$

$$F^{IV} = 2 \cos x + (4)(\cos x) + (4x)(-\sin x) + (-2x)(\sin x) + (-x^2)(\cos x)$$

$$2 \cos x + 4 \cos x - 4x \sin x - 2x \sin x - x^2 \cos x$$

$$6 \cos x - 6x \sin x - x^2 \cos x \quad F^{IV}(0) = 6$$

$$P(x) = 6 \frac{x^4}{24} + o(x^4) = \frac{x^4}{4} + o(x^4)$$

3)

Dato una funzione continua f su $[a, b]$ se $f(a)f(b) < 0$ allora esiste $c \in (a, b)$ tale che $f(c) = 0$

4)

$$F[\pi, 2\pi]$$

$$F(x) = -x^2 \cos x + 2x \sin x + 2 \cos x - 2$$

$$F(\pi) = -\pi^2 \cos \pi + 2\pi \sin \pi + 2 \cos \pi - 2$$

$$\pi^2 - 4 > 0$$

$$F(2\pi) = -2\pi \cos 2\pi + 4\pi \sin 2\pi + 2\pi - 2$$

$$-2\pi + 2\pi - 2 = -2 < 0$$

Per il teorema degli zeri esiste $c \in (\pi, 2\pi)$ tale che $F(c) = 0$

Eg 2

1)

Per il criterio delle radici per successioni, dato a_n definitivamente non negativo e $\lim \sqrt[n]{a_n} = l \in \mathbb{R}$, con $l < 1$ $a_n \rightarrow 0$ invece con $l > 1$ $a_n \rightarrow +\infty$

2)

$$a_n = \frac{(4n+1)^n}{n^{2n}+1}$$

$$\lim \frac{\sqrt[n]{(4n+1)^n}}{\sqrt[n]{n^{2n}+1}} = \frac{4n+1}{\sqrt[n]{n^{2n} \cdot n}} = \frac{4n+1}{n^{2n} \sqrt[n]{n}} = \frac{4}{n \sqrt[n]{n}} = 0^+$$

3)

Esistono che $a_n \rightarrow 0$ / $a_n \rightarrow +\infty$ fa anche così che $\sum 1/a_n$ non soddisfa le condizioni di convergenza implicando così la divergenza della serie

4)

Dal punto ii $a_n \rightarrow 0^+$ quindi $0 < a_n < 1$ ossia $(4n+1)^n < n^{2n}+1$